Partial Derivatives and Gradients

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Derivatives of one variable.
- **Leads to:** Jacobian and matrix gradients; Hessian and multivariate Taylor; Gradient descent; Lagrange multipliers, KKT and duality; Maximum likelihood estimation (MLE); Backpropagation; Fast marching and the Eikonal equation; Potential fields.

1. Overview

Key point: For a function of many inputs, we take one derivative per input with the others held fixed; collected in a row, these **partial derivatives** (G-1457) form the gradient, an arrow that points straight uphill.

This Note follows Chapter 5 (Section 5.2) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

1.1 Reading a contour map

Key point: A contour map is a surface seen from straight above. Each line joins the points where the surface has the same height.

A loss depends on many parameters at once. With two inputs we can still draw it: each pair of inputs gets a height. Our running example takes two numbers, x_1 and x_2, and gives one number:

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$$

At the point $(1, 1)$:

$$f(1, 1) = 1 + 1 + 2 = 4$$

Section 2 works through this sum term by term. Drawn in 3D, with the height f going up, the function is a bowl (Figure 1, first frame). The bowl is lowest at $(0, 0)$, where $f = 0$, and rises in every direction. It is a little tilted and stretched: it rises faster along x_2, because x_2 carries the factor 2.

A 3D bowl is hard to draw on flat paper and hard to read numbers from. A **contour map** (G-468, also called a contour plot) flattens it. Figure 1 builds one, step by step:

1. **Slice.** A flat horizontal plane (orange) cuts the bowl at one height, say $f = 5$. Where the plane meets the bowl, it leaves a ring: every point on that ring has height 5.
2. **Drop.** The ring falls straight down onto the floor. It keeps its shape and its colour.

3. **Repeat.** We slice at heights 2, 5, 8 and 11, then at every height from 0.5 to 14 in steps of 1.5. Each slice drops its own ring.
4. **Look from above.** The camera tilts up until it looks straight down. Seen from above, each ring on the bowl sits exactly over its copy on the floor. The floor is now the contour map. The black dot marks the input $(1, 1)$, with height 4, on the bowl and on the floor.

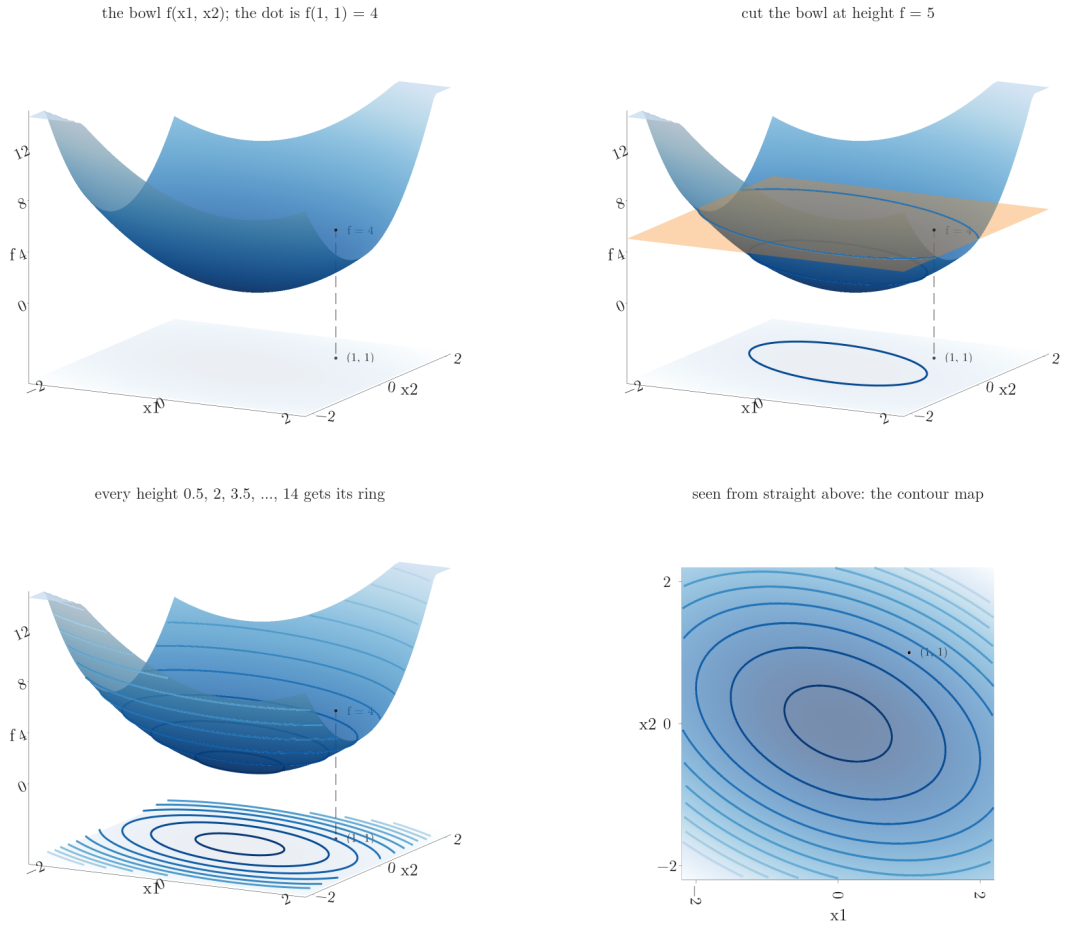


Figure 1: How a contour map is made. The bowl $f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$ is cut by horizontal planes; each cut is a ring of equal height that drops to the floor; the camera then tilts from the side to straight above, and the floor becomes the contour map of Figure 2. Darker blue = lower. Black dot: $(1, 1)$, where $f = 4$. Our own animation

Each line on a contour map is called a **contour line**: it joins points of the same height. How to read the map:

- **One line = one height.** Walk along a contour line and the height never changes. The points $(1, 1)$ and $(2, -1)$ both have height 4, so they lie on the same line (Figure 3).
- **Lines close together = steep.** Neighbouring lines differ by the same height, 1.5 here. Where the surface is steep, it climbs 1.5 in a short step, so the lines crowd together.

Where it is nearly flat, we must walk far to climb 1.5, so the lines spread apart. In Figure 1 the lines crowd near the edges, where the bowl walls are steep.

- **The innermost ring circles the lowest point.** The rings shrink towards the bottom of the bowl, at $(0, 0)$.
- **Colour shows height,** here darker = lower, as on the bowl.

1.2 The gradient on the map

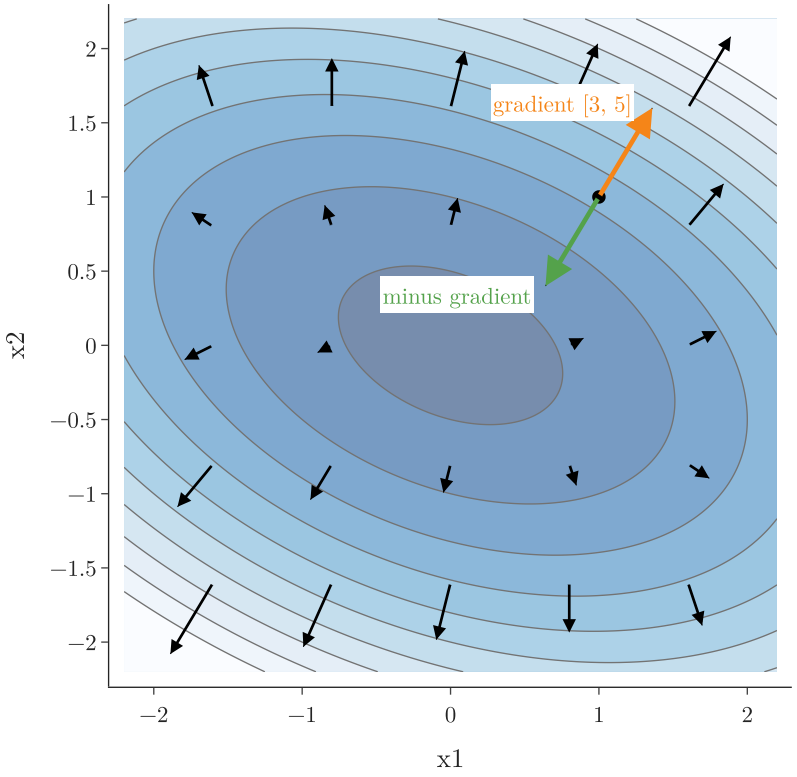


Figure 2: Contour map of $f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$ with gradient arrows: each crosses the contour lines at right angles and points uphill

Figure 2 is the contour map from the end of Figure 1, with arrows added. At every point, the **gradient** (G-863) is an arrow pointing in the steepest uphill direction. Its opposite, the green arrow, is the step **gradient descent** (G-862) takes.

Earlier Notes already used these ideas:

- the **partial derivative** (G-1457) as a slope in one variable, from the derivative of the error with respect to b;
- the gradient as the vector of partial derivatives of the loss, from both parameters together.

This Note adds the definition as a limit, the geometry of the gradient, the rules for vectors and the chain rule with several variables. It builds on the one-variable derivative.

2. Functions of several variables

> **Key point:** A function of several variables takes a list of numbers in and gives one number out, just as a loss takes all the parameters and gives one error.

Our running example is the tilted bowl of Figure 1, shaped like a loss surface. It takes two numbers, x_1 and x_2, and gives one number:

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$$

At $(1, 1)$, one term per line:

$$x_1^2 = 1^2 = 1$$

$$x_1 x_2 = 1 \times 1 = 1$$

$$2x_2^2 = 2 \times 1^2 = 2$$

$$f(1, 1) = 1 + 1 + 2 = 4$$

Such a function is a **function of several variables** (G-815). We hold its inputs together in one bold letter, the vector $\mathbf{x} = [x_1, x_2]$; at our point, $\mathbf{x} = [1, 1]$.

Reading the notation. Each symbol, with an instance:

Symbol	Read as	Instance
$\mathbb{R}$	the real numbers: any number on the number line	4, -0.5, 1.1
$\mathbb{R}^2$	all pairs of numbers	$[1, 1]$, $[2, -1]$
$\mathbb{R}^n$	all lists of n numbers; n counts the inputs	our bowl has $n = 2$; a model with 1000 parameters has $n = 1000$
$\in$	"is in", "is one of"	$[1, 1] \in \mathbb{R}^2$: the input is a pair of numbers
$f : \mathbb{R}^n \to \mathbb{R}$	f takes n numbers in and gives one number out	our bowl: $f : \mathbb{R}^2 \to \mathbb{R}$
$\mathbf{x} \mapsto f(\mathbf{x})$	"the input $\mathbf{x}$ is sent to the output $f(\mathbf{x})$"	$[1, 1] \mapsto 4$
$^\mathsf{T}$ (transpose)	turns a row into a column, and back	$[1, 2]^\mathsf{T}$ is the column with 1 on top of 2
$\mathbb{R}^{1 \times n}$	all rows of n numbers (1 row, n columns)	$[3, 5] \in \mathbb{R}^{1 \times 2}$

In general we write

$$f : \mathbb{R}^n \to \mathbb{R}$$

$$\mathbf{x} \mapsto f(\mathbf{x})$$

$$\mathbf{x} = [x_1, \ldots, x_n]^\mathsf{T}$$

The $^\mathsf{T}$ only says that the inputs are stacked as a column; the numbers are the same.

A second example we use: the squared length of a vector,

$$f(\mathbf{x}) = \mathbf{x}^\mathsf{T}\mathbf{x} = x_1^2 + x_2^2$$

$$f(3, 4) = 9 + 16 = 25$$

the dot product of $\mathbf{x}$ with itself (see computing the dot product).

Figure 3 marks four inputs of the running example on its contour map (the bowl of Figure 1 seen from above). Each input is a vector of two numbers, and each gets back a single number:

$$[1, 1] \mapsto 4$$

$$[0, 1] \mapsto 2$$

$$[-1, 0] \mapsto 1$$

Two different inputs can give the same number: $[2, -1]$ also gives 4, so it lies on the same contour line as $[1, 1]$.

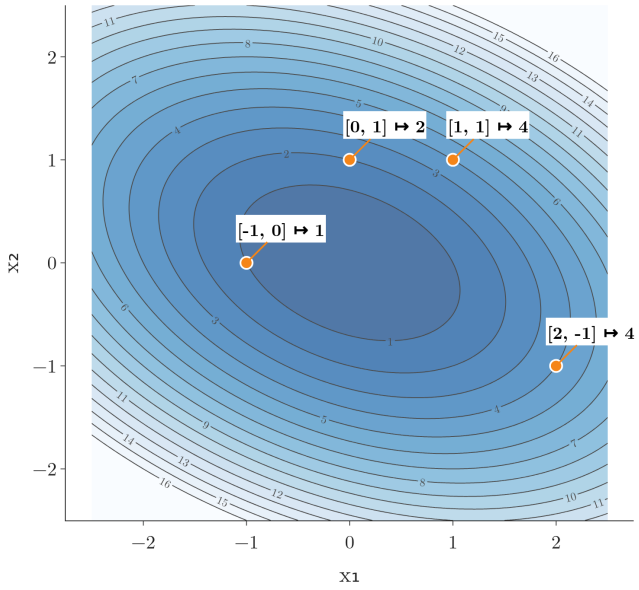


Figure 3: The running example on a contour map, with four input vectors and the single number the function returns for each. [1, 1] and [2, −1] both give 4 and lie on the same contour line.

3. Partial derivatives

> **Key point:** A **partial derivative** (G-1457) changes one input by a tiny step, keeps all the others fixed, and measures the slope; it is an ordinary one-variable derivative.

3.1 The definition

> **Key point:** Nudge only x_i by h, take the difference quotient, and let $h \to 0$.

A one-variable derivative answers: if we nudge the input a little, how many times bigger is the change in the output? With two inputs the question needs one more detail: nudge **which** input? Figure 4 tries both on our bowl at $(1, 1)$, where $f = 4$.

1. Nudge only x_1 by 0.1. The output moves from 4 to $f(1.1, 1) = 4.31$: a change of 0.31, about 3 times the nudge.
2. Go back, and nudge only x_2 by 0.1. The output moves from 4 to $f(1, 1.1) = 4.52$: a change of 0.52, about 5 times the nudge.

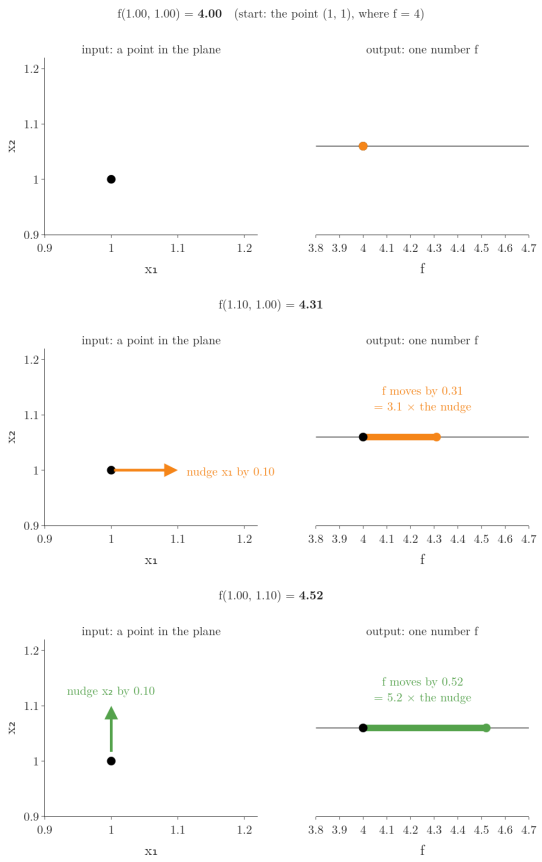


Figure 4: The point $(1, 1)$ in the input plane (left) and the single output number f on a number line (right). A nudge of 0.1 in x_1 alone moves the output by 0.31; the same nudge in x_2 alone moves it by 0.52. Idea after Khan Academy, "Partial derivatives, introduction"

The same function responds differently to each input. Each of the two ratios tells only part of the story of how f changes, which is where the name comes from. The derivative in the error function was the first **partial derivative** (G-1457) as the slope of a **function of several variables** (G-815) in one variable, holding the others fixed. Written with the limit of the one-variable derivative (a limit of the difference quotient, the slope of a secant line):

1. **In words:** move only x_i by a step h, divide the change in f by h, and let h shrink to 0.
2. **Formula:**
$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(\mathbf{x})}{h}$$
The curly ∂ (read "partial") replaces d to say that the other variables are held fixed.
3. **Example:** for our bowl at $(1, 1)$ with $h = 0.1$,
$$f(1.1,\ 1) = 1.1^2 + 1.1 \times 1 + 2 \times 1^2$$
$$f(1.1,\ 1) = 1.21 + 1.1 + 2 = 4.31$$
$$f(1.1,\ 1) - f(1, 1) = 4.31 - 4 = 0.31$$

$$\frac{0.31}{0.1} = 3.1$$

Smaller h brings this towards 3.

3.2 Computing partial derivatives

> **Key point:** Treat every other variable as a constant and use the one-variable rules.

Because the other variables are frozen, they behave like numbers. All the rules for computing derivatives apply unchanged.

1. **In words:** to differentiate with respect to x_1, read x_2 as a fixed number, and the other way round.
2. **Formula:** for our bowl
$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$$
the **partial derivatives** (G-1457) are
$$\frac{\partial f}{\partial x_1} = 2x_1 + x_2$$
$$\frac{\partial f}{\partial x_2} = x_1 + 4x_2$$
In $\partial f/\partial x_1$, the term $x_1 x_2$ is x_1 times a constant, so its derivative is that constant x_2; the term $2x_2^2$ is a constant, so its derivative is 0.
3. **Example:** at $(1, 1)$:
$$\frac{\partial f}{\partial x_1} = 2 \times 1 + 1 = 3$$
$$\frac{\partial f}{\partial x_2} = 1 + 4 \times 1 = 5$$

The **chain rule** (G-371) works the same way inside a partial derivative. Take

$$g(x, y) = (x + y^2)^2$$

$$g(1, 2) = (1 + 4)^2 = 25$$

The inside is $u = x + y^2$, which is 5 at $(1, 2)$. The outside is u^2, with derivative $2u = 10$. Then multiply by how the inside responds to each input:

$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial u}{\partial y} = 2y = 4$$

$$\frac{\partial g}{\partial x} = 2u \times \frac{\partial u}{\partial x} = 10 \times 1 = 10$$

$$\frac{\partial g}{\partial y} = 2u \times \frac{\partial u}{\partial y} = 10 \times 4 = 40$$

3.3 The picture: slopes of two slices

> **Key point:** Fixing x_2 cuts the surface along a curve in the x_1 direction; $\partial f/\partial x_1$ is the slope of its tangent line.

Figure 5 shows the surface of our bowl. Fixing $x_2 = 1$ cuts it along the orange curve, a function of x_1 alone. Its tangent line at $(1, 1)$ has slope 3: that is $\partial f/\partial x_1$. Fixing $x_1 = 1$ instead gives the green curve, whose tangent line has slope 5: that is $\partial f/\partial x_2$.

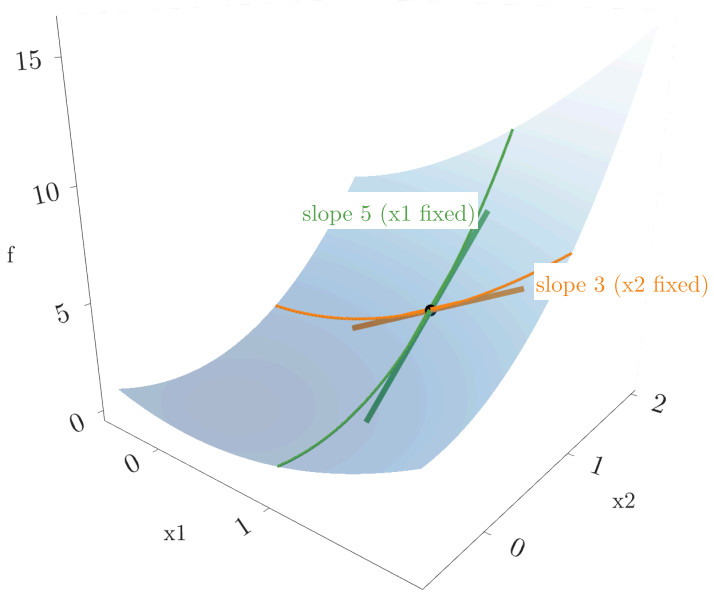


Figure 5: The surface $f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$ cut along $x_2 = 1$ (orange) and along $x_1 = 1$ (green); the thick lines are the tangent lines at $(1, 1)$

The slice picture needs a surface, so it only works for two inputs. The nudge picture of Figure 4 needs no surface: nudge one input, watch the one output number. That reading still works for a loss with a million parameters.

The two tangent lines together span a flat plane that touches the surface at $(1, 1)$, the tangent plane. Linearisation uses it to approximate the surface.

4. The gradient

> **Key point:** The gradient collects all **partial derivatives** (G-1457) in one row vector; as an arrow, it points in the direction of steepest ascent and is perpendicular to the contour lines.

4.1 Collecting the partial derivatives

> **Key point:** One **partial derivative** (G-1457) per input, side by side: a row of n numbers (a $1 \times n$ row vector).

Gradient descent (G-862) (both parameters together) already used the **gradient** as the vector of partial derivatives of the loss, pointing in the direction of steepest increase. We write it as a row:

1. **In words:** list the partial derivatives in the order of the inputs.
2. **Formula:**
$$\nabla_{\mathbf{x}} f = \frac{df}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{1\times n}$$
∇ is read "nabla" (G-1295) or "grad". The tag $\in \mathbb{R}^{1\times n}$ says the result is one row of n numbers. Written as a row, this is the **gradient as a row vector** (G-858).
3. **Example:** for our bowl,
$$\nabla f = \begin{bmatrix} 2x_1 + x_2 & x_1 + 4x_2 \end{bmatrix}$$
$$\nabla f(1, 1) = \begin{bmatrix} 3 & 5 \end{bmatrix}$$
$$\nabla f(2, -1) = \begin{bmatrix} 3 & -2 \end{bmatrix}$$

Row or column? Many texts, and the update rules of gradient descent, treat the gradient as a column vector like the parameters. The book we follow uses a row: then the **chain rule** (G-371) becomes ordinary matrix multiplication with no transposes (Section 6), and the next Note's **Jacobian** (G-980) follows naturally. The numbers are the same either way; only the shape differs. In NumPy both are plain 1-D arrays, so code rarely notices.

4.2 The gradient as an arrow on the contour map

> **Key point:** Drawn at its point, the **gradient** (G-863) arrow points the way f rises fastest, crosses the contour line at a right angle, and is longer where the surface is steeper.

Figure 2 draws the gradient as an arrow at many points of the contour map. Three things are visible:

- **It points uphill.** Every arrow points away from the minimum, towards higher contour lines.
- **It crosses contour lines at right angles.** Along a contour line f does not change at all, so the steepest direction is straight across it.

- **Its length is the steepness.** Arrows are long where contour lines are packed close together, and short near the flat bottom.

The hiker. Think of the surface as a hillside and stand on it. Which way should we walk to climb fastest? Figure 6 answers by doing it. The contour map lies under the bowl, and the orange gradient arrow is drawn on the map under the hiker. At every step the hiker reads the arrow and walks that way. The path heads straight away from the lowest point and gets steeper: the steepness readout, the length of the gradient, grows from 0.9 to 6.5.

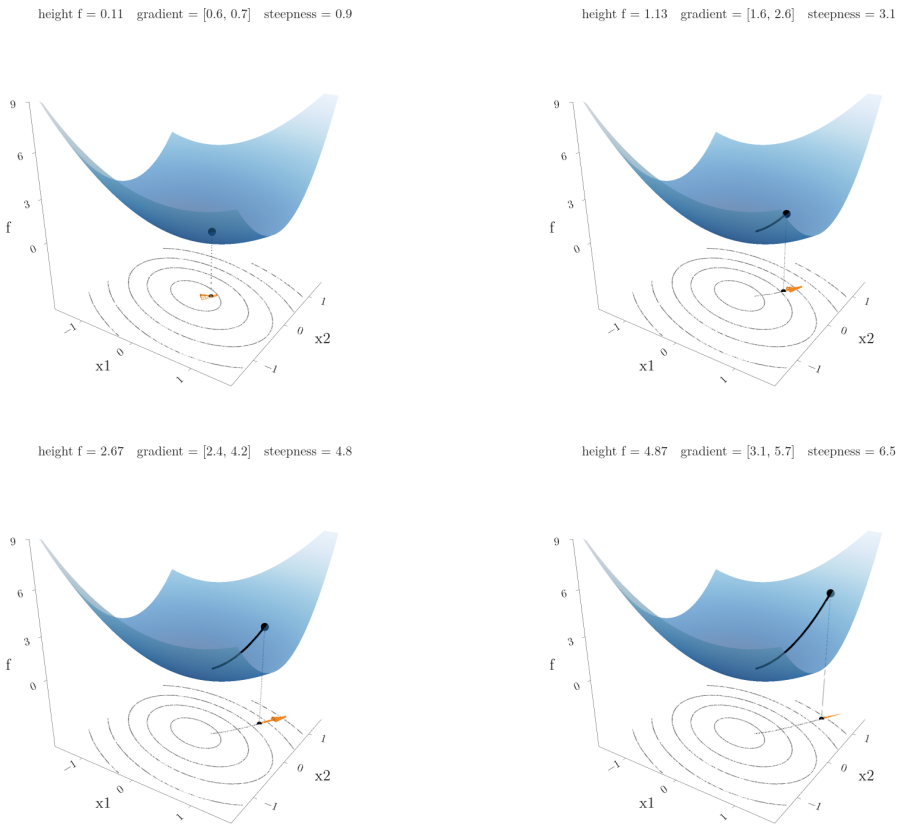


Figure 6: A hiker (black dot) climbs the bowl by always stepping along the gradient arrow (orange) drawn on the contour map below. The dashed shadow of the path crosses every contour line at a right angle, and the arrow grows as the surface gets steeper. Idea after Khan Academy, "Gradient and graphs"; our own function

Why at right angles? Figure 7 zooms in on $(1, 1)$ on the contour map of Figure 1 until the contour lines $f = 4$ and $f = 4.1$ look like two straight, parallel lines.

1. Every arrow from the first line to the second raises f by the same amount, 0.1.
2. The arrows have different lengths: 0.024, 0.019, 0.017, 0.019, 0.024.
3. The steepest direction gains that 0.1 over the shortest distance.
4. The shortest path between two parallel lines goes straight across, at a right angle to both.

So the steepest direction is perpendicular to the contour line, and it is the direction of the gradient $[3, 5]$. The shortest length also checks the third bullet: 0.1 of height over a distance of 0.017 is a slope of $0.1/0.017 \approx 5.83$, the length of the gradient, $\sqrt{3^2 + 5^2}$.

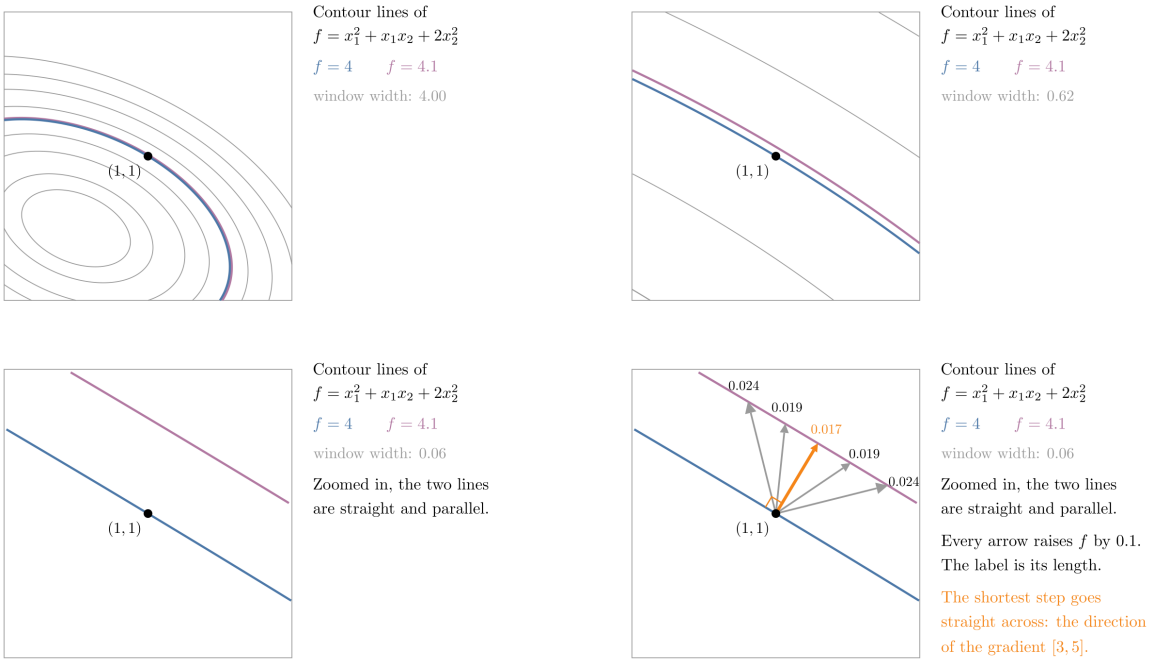


Figure 7: Zooming in on $(1, 1)$: the contour lines $f = 4$ (blue) and $f = 4.1$ (purple) become straight and parallel. Of the arrows that raise f by 0.1, the shortest (orange, length 0.017) crosses at a right angle and points along the gradient $[3, 5]$. Idea after Khan Academy, "Gradient and contour maps"; our own function

At $(1, 1)$ the gradient $[3, 5]$ (orange) points up and to the right. Its opposite, $[-3, -5]$ (green), points into the bowl: the direction **gradient descent** (G-862) steps in, $\mathbf{x}_{\text{new}} = \mathbf{x} - \eta \nabla f(\mathbf{x})$.

4.3 The slope in any direction

> **Key point:** The slope of f along a unit vector $\mathbf{u}$ is the dot product $\nabla f \cdot \mathbf{u}$; it is largest when $\mathbf{u}$ points along the **gradient** (G-863).

The **partial derivatives** (G-1457) give the slope along the two axes. A hiker can walk in any direction, so we need the slope along any arrow.

1. **In words:** a step in the direction $\mathbf{u} = [a, b]$ is made of two parts: a along x_1 and b along x_2. Each part changes f on its own:
   - the x_1 part changes f by its size a times the rate $\partial f/\partial x_1$;
   - the x_2 part changes f by its size b times the rate $\partial f/\partial x_2$.

   The slope along $\mathbf{u}$ is these two products added.

2. **Example first,** at $(1, 1)$, where $\partial f/\partial x_1 = 3$ and $\partial f/\partial x_2 = 5$. Take the direction $\mathbf{u} = [0.6, 0.8]$. It is a unit vector: its length is

$$\sqrt{0.6^2 + 0.8^2} = \sqrt{0.36 + 0.64}$$

$$\sqrt{0.36 + 0.64} = \sqrt{1} = 1$$

$$x_1 \text{ part:} \quad 0.6 \times 3 = 1.8$$

$$x_2 \text{ part:} \quad 0.8 \times 5 = 4.0$$

$$\text{slope along } \mathbf{u} = 1.8 + 4.0 = 5.8$$

3. **Formula:** two products added is the dot product of $\mathbf{u}$ with the gradient. For a unit vector $\mathbf{u} = [a, b]$ (length 1), the **directional derivative** (G-614) is

$$\nabla_{\mathbf{u}} f = a \frac{\partial f}{\partial x_1} + b \frac{\partial f}{\partial x_2}$$

$$\nabla_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

Check with the example:

$$0.6 \times 3 + 0.8 \times 5 = 5.8$$

4. **Three more directions** at $(1, 1)$, where $\nabla f = [3, 5]$. The length of the gradient is

$$\sqrt{3^2 + 5^2} = \sqrt{34} = 5.83$$

- Along $\mathbf{u} = [1, 0]$:

$$1 \times 3 + 0 \times 5 = 3$$

the partial derivative $\partial f/\partial x_1$.
- Along the gradient itself, the unit vector

$$\mathbf{u} = \frac{[3, 5]}{\sqrt{34}}$$

$$\frac{3 \times 3 + 5 \times 5}{\sqrt{34}} = \frac{9 + 25}{\sqrt{34}}$$

$$\frac{9 + 25}{\sqrt{34}} = \frac{34}{\sqrt{34}} = \sqrt{34} = 5.83$$

the largest possible.
- At right angles to the gradient, along

$$\mathbf{u} = \frac{[5, -3]}{\sqrt{34}}$$

$$\frac{3 \times 5 + 5 \times (-3)}{\sqrt{34}} = \frac{15 - 15}{\sqrt{34}} = 0$$

we walk along the contour line.

A dot product with a unit vector is the length of the gradient times the cosine of the angle between the two, so it is largest when the two point the same way (see the angle between two vectors). This alignment is why the gradient is the direction of **steepest ascent** (G-1888), and why its length, 5.83, is the slope in that direction: the same number Figure 7 found from the shortest arrow.

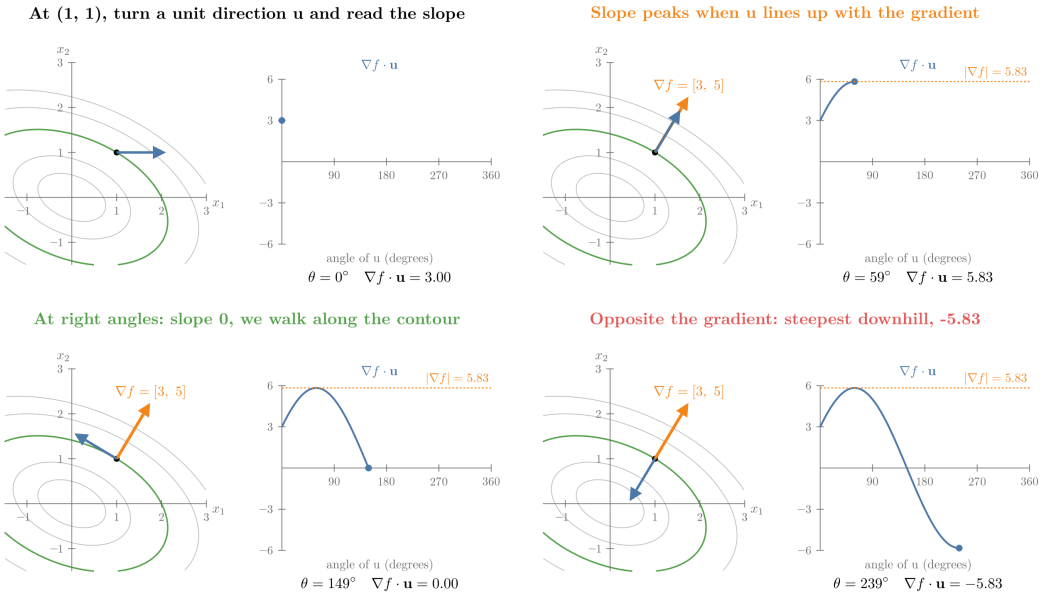


Figure 8: At $(1, 1)$ a unit direction $\mathbf{u}$ turns full circle (left); the slope $\nabla f \cdot \mathbf{u}$ is traced against its angle (right): 3 along the x_1-axis, a peak of 5.83 when $\mathbf{u}$ lines up with $\nabla f = [3, 5]$, 0 along the contour line, -5.83 straight downhill

Figure 8 turns the three directions above into one sweep. Watch the right panel. Its horizontal axis is the angle of $\mathbf{u}$ measured from the x_1-axis, the θ printed under each frame; the gradient $[3, 5]$ itself sits at $\theta = 59°$. The slope rises and falls like a cosine wave, $\sqrt{34}\cos(\theta - 59°)$: it depends only on the angle between $\mathbf{u}$ and the gradient, $\theta - 59°$. Its top sits exactly where the blue arrow covers the orange gradient arrow, and it crosses 0 where the blue arrow runs along the green contour line. The picture follows Sanderson's Khan Academy lesson on why the gradient is the direction of steepest ascent, redrawn with our function.

5. Rules for gradients

> **Key point:** Sum, product and **chain rules** (G-371) still hold with vectors, but the factors are now vectors and matrices, so their order matters.

The rules of the one-variable case carry over. We need one building block first: the **gradient** (G-863) of a dot product with a fixed vector. Take $\mathbf{a} = [1, 2]$, so

$$\mathbf{a}^\mathsf{T}\mathbf{x} = 1 \times x_1 + 2 \times x_2$$

Its **partial derivatives** (G-1457) are just the two fixed numbers:

$$\frac{\partial}{\partial x_1}(x_1 + 2x_2) = 1$$

$$\frac{\partial}{\partial x_2}(x_1 + 2x_2) = 2$$

So the gradient is $\mathbf{a}^\mathsf{T} = [1, 2]$. In general,

$$\frac{\partial}{\partial \mathbf{x}}\, \mathbf{a}^\mathsf{T}\mathbf{x} = \mathbf{a}^\mathsf{T}$$

Now the three rules, each for two functions f and g that take n numbers in and give one number out ($f, g : \mathbb{R}^n \to \mathbb{R}$), and each with numbers at $\mathbf{x} = (1, 1)$.

Sum rule.

1. **In words:** the gradient of a sum is the sum of the gradients, entry by entry.
2. **Example:** f is our bowl and g is the dot product with $\mathbf{a} = [1, 2]$:

   $$g(\mathbf{x}) = \mathbf{a}^\mathsf{T}\mathbf{x}$$

   $$g(\mathbf{x}) = x_1 + 2x_2$$

   At $(1, 1)$:

   $$\nabla f = [3,\ 5]$$
   $$\nabla g = [1,\ 2]$$
   $$\nabla(f + g) = [3 + 1,\ 5 + 2]$$
   $$\nabla(f + g) = [4,\ 7]$$

3. **Formula:**

   $$\frac{\partial}{\partial \mathbf{x}}\Big(f(\mathbf{x}) + g(\mathbf{x})\Big) = \frac{\partial f}{\partial \mathbf{x}} + \frac{\partial g}{\partial \mathbf{x}}$$

Product rule.

1. **In words:** differentiate one factor at a time, keep the other, add.
2. **Example:** $\mathbf{a} = [1, 2]$, $\mathbf{b} = [3, -1]$, and the product of two dot products

   $$f(\mathbf{x}) = (\mathbf{a}^\mathsf{T}\mathbf{x})(\mathbf{b}^\mathsf{T}\mathbf{x}) = (x_1 + 2x_2)(3x_1 - x_2)$$

   At $\mathbf{x} = [1, 1]$ the two factors are

   $$\mathbf{a}^\mathsf{T}\mathbf{x} = 1 + 2 = 3$$

   $$\mathbf{b}^\mathsf{T}\mathbf{x} = 3 - 1 = 2$$

First term, the gradient of the first factor times the value of the second:

$$2 \times [1,\ 2] = [2,\ 4]$$

Second term, the value of the first factor times the gradient of the second:

$$3 \times [3,\ -1] = [9,\ -3]$$

Add:

$$\nabla f = [2 + 9,\ 4 - 3]$$
$$\nabla f = [11,\ 1]$$

Check by multiplying out first:

$$f = 3x_1^2 + 5x_1x_2 - 2x_2^2$$

Then:

$$\frac{\partial f}{\partial x_1} = 6x_1 + 5x_2 = 6 + 5 = 11$$
$$\frac{\partial f}{\partial x_2} = 5x_1 - 4x_2 = 5 - 4 = 1$$

3. **Formula:**
$$\frac{\partial}{\partial \mathbf{x}}\Big(f(\mathbf{x})\, g(\mathbf{x})\Big) = \frac{\partial f}{\partial \mathbf{x}}\, g(\mathbf{x}) + f(\mathbf{x})\, \frac{\partial g}{\partial \mathbf{x}}$$
For the example: $\nabla f = (\mathbf{b}^\mathsf{T}\mathbf{x})\, \mathbf{a}^\mathsf{T} + (\mathbf{a}^\mathsf{T}\mathbf{x})\, \mathbf{b}^\mathsf{T}$.

Chain rule (one outer function of one number).

1. **In words:** the derivative of the outer function, taken at the inner value, times the gradient of the inner function.
2. **Example:** square the bowl, $h(\mathbf{x}) = g(f(\mathbf{x}))$ with $g(u) = u^2$. At $(1, 1)$:

$$f(1, 1) = 4$$

$$g'(u) = 2u = 2 \times 4 = 8$$
$$\nabla f = [3,\ 5]$$
$$\nabla h = 8 \times [3,\ 5] = [24,\ 40]$$

3. **Formula:**
$$\frac{\partial}{\partial \mathbf{x}}\, g\Big(f(\mathbf{x})\Big) = \frac{\partial g}{\partial f}\, \frac{\partial f}{\partial \mathbf{x}}$$

With vectors and matrices, AB and BA are different things (see order matters). So we keep the order written above and check that the shapes fit.

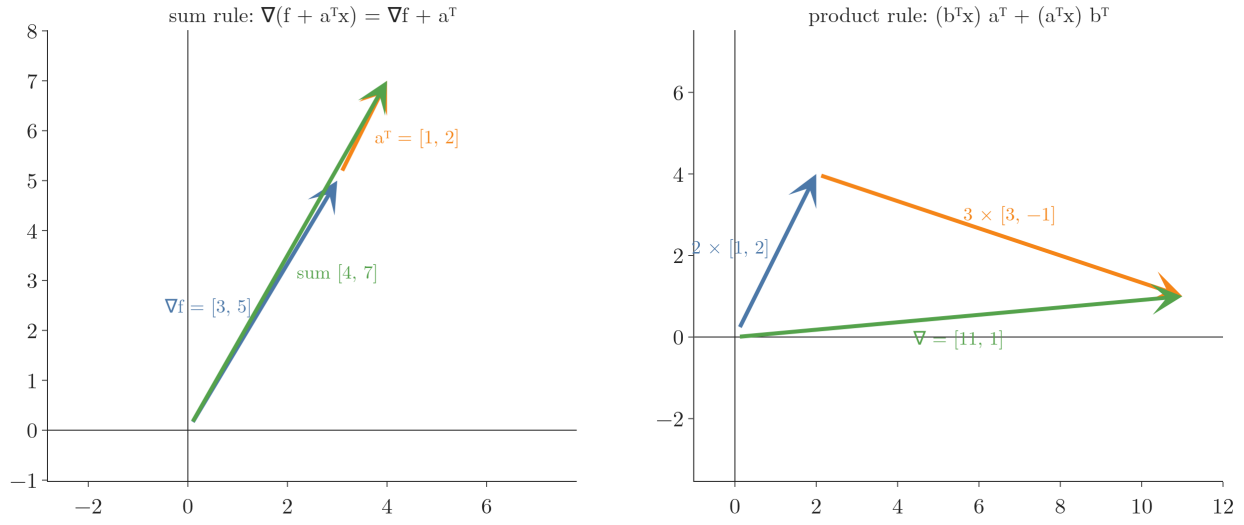


Figure 9: Gradient rules as additions of arrows at $\mathbf{x} = (1, 1)$. Left, sum rule: the gradient of $f + \mathbf{a}^\mathsf{T}\mathbf{x}$ is $\nabla f = [3, 5]$ plus $\mathbf{a}^\mathsf{T} = [1, 2]$, giving $[4, 7]$. Right, product rule: $2 \times [1, 2]$ plus $3 \times [3, -1]$ gives the gradient $[11, 1]$.

Figure 9 draws both rules as arrows added tip to tail. A gradient is a vector, so the sum rule adds two arrows. The product rule adds two arrows too, each one gradient scaled by the value of the other factor: $\mathbf{a}^\mathsf{T}$ scaled by 2 and $\mathbf{b}^\mathsf{T}$ scaled by 3.

6. The chain rule with several variables

> **Key point:** When an input affects f through several intermediate variables, we multiply along each path and add up the paths; written with **gradients** (G-863), this is a matrix product.

This rule is the **multivariate chain rule** (G-1281), the several-variable form of the **chain rule** (G-371).

6.1 One outer input, two paths

> **Key point:** The rate of change of f with t is the sum over both paths of (slope of f along the path) times (rate of the intermediate variable): a row vector times a column vector.

Suppose the inputs x_1 and x_2 are themselves functions of one variable t. For example, a point moves around a circle of radius 1:

$$x_1(t) = \cos t$$

$$x_2(t) = \sin t$$

At $t = 0$ the point is $(\cos 0, \sin 0) = (1, 0)$; at $t = \pi/2$ it is $(\cos \frac{\pi}{2}, \sin \frac{\pi}{2}) = (0, 1)$. The function is our bowl,

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$$

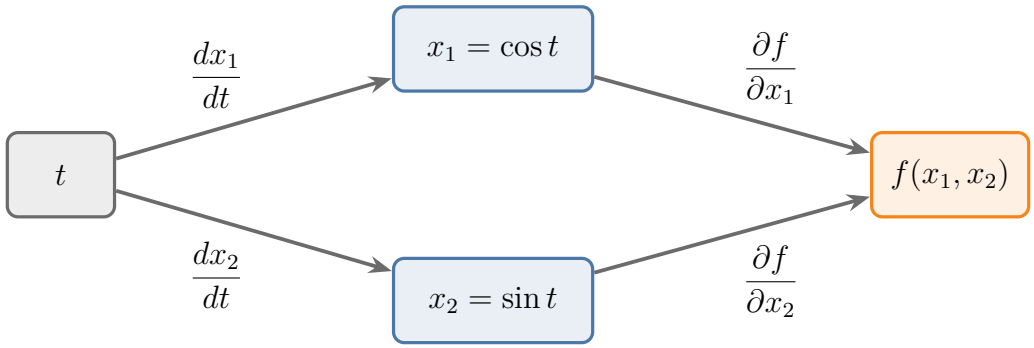
$$f(1, 0) = 1$$

$$f(0, 1) = 2$$

How fast does the value of f change as t changes? A change in t reaches f along two paths (Figure 10): through x_1 and through x_2.

How to read Figure 10:

- each box is one quantity: t, the two inputs x_1 and x_2, and the output f;
- an arrow from one box to the next means "the next one is computed from this one";
- the label on an arrow is a derivative: how fast the box at the head changes when the box at the tail changes;
- the formula at the bottom multiplies the labels along each path, then adds the two paths. The steps below do this with numbers.



$$\frac{df}{dt} = \underbrace{\frac{\partial f}{\partial x_1}\frac{dx_1}{dt}}_{\text{path through } x_1} + \underbrace{\frac{\partial f}{\partial x_2}\frac{dx_2}{dt}}_{\text{path through } x_2}$$

Figure 10: The multivariate chain rule: multiply the derivatives along each path from t to f, then add the paths

Figure 11 shows the same thing as a process. Its left panel is the contour map of the bowl of Figure 1, seen from above; darker lines are lower, as on the bowl of Figure 1 (the panel's title

says so). As t grows, the point goes round the circle, and its step splits into a part along x_1 and a part along x_2. Each part changes f on its own; the right panel shows the total.

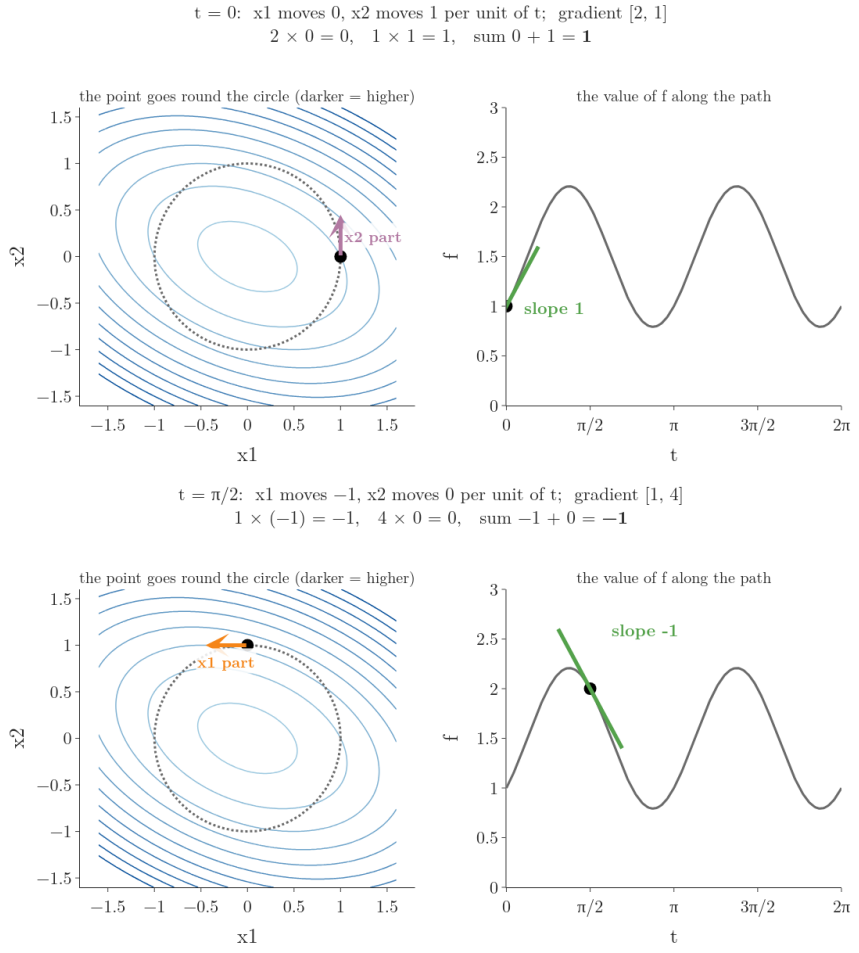


Figure 11: A point goes round the unit circle on the contour map of the bowl (left); the right panel tracks the value of f. At $t = 0$ the step is all along x_2 (purple) and f rises with slope 1; at $t = \pi/2$ it is all along x_1 (orange) and f falls with slope -1. Idea after Khan Academy, "Multivariable chain rule" and "Multivariable chain rule intuition"; our own function

The steps at $t = \pi/2$, where the point is $(0, 1)$.

1. **How fast each input moves.** The derivatives of $\cos t$ and $\sin t$ are $-\sin t$ and $\cos t$:

$$\frac{dx_1}{dt} = -\sin\tfrac{\pi}{2} = -1$$

$$\frac{dx_2}{dt} = \cos\tfrac{\pi}{2} = 0$$

2. **How strongly f responds to each input** at $(0, 1)$, from Section 3.2:

$$\frac{\partial f}{\partial x_1} = 2x_1 + x_2 = 0 + 1 = 1$$

$$\frac{\partial f}{\partial x_2} = x_1 + 4x_2 = 0 + 4 = 4$$

3. **One product per path:**

$$\text{through } x_1: \quad 1 \times (-1) = -1$$

$$\text{through } x_2: \quad 4 \times 0 = 0$$

4. **Add the paths:**

$$\frac{df}{dt} = -1 + 0 = -1$$

The same steps at $t = 0$, where the point is $(1, 0)$:

$$\frac{dx_1}{dt} = -\sin 0 = 0$$

$$\frac{dx_2}{dt} = \cos 0 = 1$$

$$\frac{\partial f}{\partial x_1} = 2 + 0 = 2$$

$$\frac{\partial f}{\partial x_2} = 1 + 0 = 1$$

$$\text{through } x_1: \quad 2 \times 0 = 0$$

$$\text{through } x_2: \quad 1 \times 1 = 1$$

$$\frac{df}{dt} = 0 + 1 = 1$$

These are the slopes of the green tangent lines in Figure 11.

The rule, in general:

1. **In words:** for each intermediate variable, multiply "how f responds to it" by "how it responds to t"; add the results.
2. **Formula:**

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1}\frac{dx_1}{dt} + \frac{\partial f}{\partial x_2}\frac{dx_2}{dt}$$

The same sum, written as a product of two vectors:

$$\frac{df}{dt} = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} & \dfrac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \dfrac{dx_1}{dt} \\ \dfrac{dx_2}{dt} \end{bmatrix}$$

The second form is the **gradient** (G-863) (a row) times the velocity of the point (a column). At $t = \pi/2$:

$$\begin{bmatrix} 1 & 4 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$
$$= 1 \times (-1) + 4 \times 0$$
$$= -1$$

Check the long way, without the rule. Put the circle into the bowl first, so that f becomes a function of t alone:

$$f(t) = \cos^2 t + \cos t \sin t + 2 \sin^2 t$$

Use $\cos^2 t + \sin^2 t = 1$ to replace $\cos^2 t + \sin^2 t$ inside it:

$$f(t) = 1 + \cos t \sin t + \sin^2 t$$

Use $\cos t \sin t = \frac{1}{2} \sin 2t$:

$$f(t) = 1 + \sin^2 t + \tfrac{1}{2} \sin 2t$$

Differentiate term by term with the one-variable rules:

$$\frac{d}{dt} \sin^2 t = 2 \sin t \cos t = \sin 2t$$

$$\frac{d}{dt} \tfrac{1}{2} \sin 2t = \tfrac{1}{2} \times 2 \cos 2t = \cos 2t$$

$$\frac{df}{dt} = \sin 2t + \cos 2t$$

At the two points:

$$t = \tfrac{\pi}{2} : \quad \sin \pi + \cos \pi = 0 + (-1) = -1$$

$$t = 0 : \quad \sin 0 + \cos 0 = 0 + 1 = 1$$

The same two answers. The long way needs a new round of algebra for every function. The **chain rule** (G-371) gives the same result from two small pieces that are already known, the gradient and the velocity.

The second form is a 1×2 row times a 2×1 column, giving a 1×1 number. The chain rule is where the row-vector convention pays off: the gradient sits on the left and the shapes fit with no transposing.

6.2 Two outer inputs: a matrix of partial derivatives

> **Key point:** With inputs s and t, the inner derivatives fill a 2×2 matrix, and the **chain rule** (G-371) becomes (row gradient) times (matrix).

Now let x_1 and x_2 depend on two variables s and t, for example $x_1 = s + t$ and $x_2 = st$ (at $s = 1$, $t = 2$ these are 3 and 2). Each of $\partial f/\partial s$ and $\partial f/\partial t$ follows the path rule:

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial s}$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t}$$

1. **In words:** both equations at once are one row vector times one matrix.
2. **Formula:** The left side, $df/d(s,t)$, is the row of the two answers, $\partial f/\partial s$ and $\partial f/\partial t$:

$$\frac{df}{d(s,t)} = \underbrace{\begin{bmatrix} \dfrac{\partial f}{\partial x_1} & \dfrac{\partial f}{\partial x_2} \end{bmatrix}}_{\partial f/\partial \mathbf{x}} \underbrace{\begin{bmatrix} \dfrac{\partial x_1}{\partial s} & \dfrac{\partial x_1}{\partial t} \\ \dfrac{\partial x_2}{\partial s} & \dfrac{\partial x_2}{\partial t} \end{bmatrix}}_{\partial \mathbf{x}/\partial(s,t)}$$

3. **Example:** our bowl with $x_1 = s + t$ and $x_2 = st$, at $s = 1$, $t = 2$. The inner values:

$$x_1 = 1 + 2 = 3$$

$$x_2 = 1 \times 2 = 2$$

   The **gradient** (G-863) of the bowl there:

$$\frac{\partial f}{\partial x_1} = 2 \times 3 + 2 = 8$$

$$\frac{\partial f}{\partial x_2} = 3 + 4 \times 2 = 11$$

   The inner derivatives:

$$\frac{\partial x_1}{\partial s} = 1$$

$$\frac{\partial x_1}{\partial t} = 1$$

$$\frac{\partial x_2}{\partial s} = t = 2$$

$$\frac{\partial x_2}{\partial t} = s = 1$$

   The row times the matrix, one column at a time. Column 1 (for s):

$$8 \times 1 + 11 \times 2 = 8 + 22 = 30$$

Column 2 (for t):

$$8 \times 1 + 11 \times 1 = 8 + 11 = 19$$

$$\begin{bmatrix} 8 & 11 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 30 & 19 \end{bmatrix}$$

The matrix in the middle is the first example of a **Jacobian** (G-980), the subject of the Jacobian. The chain rule as a product of matrices mirrors **composition** (G-431): doing one function after another multiplies their matrices.

The notation also reads like fractions: $\partial \mathbf{x}$ appears "below" in the first factor and "above" in the second, and "cancels" to leave $\partial f / \partial (s, t)$. The fraction reading is a memory aid only; **partial derivatives** (G-1457) are not really fractions.

7. Checking a gradient numerically

> **Key point:** Nudge each input by a small h and compare the finite-difference slope with the formula; a tiny relative error means the formula and code are probably right.

Hand-derived gradients are easy to get wrong, and a wrong gradient makes training fail quietly. The definition of the **partial derivative** (G-1457) gives a simple test, called **gradient checking** (G-860):

1. **In words:** estimate each partial derivative with a small step h (say 10^{-4}), then compare the whole estimated **gradient** (G-863) with the formula's.
2. **Example first.** The squared-error loss of **gradient descent** (G-862) on three points, inputs $x = [1, 2, 3]$ and targets $y = [2, 4, 5]$. A line with slope m and intercept b predicts $mx_i + b$ for point i; the loss adds up the squared misses ($\sum_i$ means "add over the points $i = 1, 2, 3$"):

$$L(m, b) = \sum_i (y_i - mx_i - b)^2$$

At $m = 1$, $b = 0$ the predictions are $1, 2, 3$, and the misses (the residuals r_i) are

$$r = [2 - 1,\ 4 - 2,\ 5 - 3] = [1,\ 2,\ 2]$$

The formula for the slope's partial derivative, one product per point:

$$r_1 x_1 = 1 \times 1 = 1$$

$$r_2 x_2 = 2 \times 2 = 4$$

$$r_3 x_3 = 2 \times 3 = 6$$

$$\frac{\partial L}{\partial m} = -2 \sum_i r_i x_i$$

$$\frac{\partial L}{\partial m} = -2 \times (1 + 4 + 6) = -22$$

And for the intercept:

$$\frac{\partial L}{\partial b} = -2\sum_i r_i$$

$$\frac{\partial L}{\partial b} = -2 \times (1 + 2 + 2) = -10$$

A **central difference** takes the slope between the points a step h to either side. Here $\mathbf{p} = [m, b]$ holds the parameters and $\mathbf{e}$ is $[1, 0]$ (nudge m) or $[0, 1]$ (nudge b):

$$\frac{L(\mathbf{p} + h\mathbf{e}) - L(\mathbf{p} - h\mathbf{e})}{2h}$$

With $h = 10^{-4}$, the code below gives $[-22.0000, -10.0000]$.

3. **Formula:** call d_i the formula's value for input i (here $d_1 = -22$, $d_2 = -10$) and d_i^h the finite-difference estimate with step h. Then

$$\text{relative error} = \sqrt{\frac{\sum_i (d_i^h - d_i)^2}{\sum_i (d_i^h + d_i)^2}}$$

A value below about 10^{-6} means the formula is very probably correct (MML §5.2). Here the code reports about 10^{-13}.

Python:

```
import numpy as np

x = np.array([1.0, 2.0, 3.0])
y = np.array([2.0, 4.0, 5.0])

def loss(p):
    m, b = p
    return np.sum((y - m * x - b) ** 2)

def grad(p):
    r = y - p[0] * x - p[1]
    return np.array([-2 * np.sum(r * x), -2 * np.sum(r)])

p, h = np.array([1.0, 0.0]), 1e-4
num = np.array([(loss(p + h * e) - loss(p - h * e)) / (2 * h)
                for e in np.eye(2)])
rel = np.linalg.norm(num - grad(p)) / np.linalg.norm(num + grad(p))
num, grad(p), rel      # [-22, -10], [-22, -10], about 2.5e-13
```

`np.eye(2)` gives the rows $[1, 0]$ and $[0, 1]$, so `p + h * e` nudges one parameter at a time.

Figure 12 shows how the size of the step h affects the check on this loss. How to read it:

- **Both axes use a log scale.** Each labelled tick is 100 times the one before it (10^{-8}, 10^{-6}, 10^{-4}, ...). A log scale fits very small and very large numbers on one picture.
- **Lower is better.** The height of a dot is the relative error of the check for that step h.

- **The dashed line is the pass mark,** a relative error of 10^{-6}.

The one-sided difference steps to one side only:

$$\frac{L(\mathbf{p} + h\mathbf{e}) - L(\mathbf{p})}{h}$$

It is off by an amount that shrinks in proportion to h; it only passes the 10^{-6} line for h below about 3×10^{-6}. The central difference has no such error on this loss: the loss is quadratic, and for a quadratic the central difference is exact. The reason, step by step: along one nudge direction a quadratic loss reads as a parabola in h, with numbers a, s (the true slope) and c:

$$L(\mathbf{p} + h\mathbf{e}) = a + sh + ch^2$$

$$L(\mathbf{p} - h\mathbf{e}) = a - sh + ch^2$$

$$\text{difference} = 2sh$$

$$\frac{2sh}{2h} = s$$

The a and ch^2 terms cancel, so the estimate is the true slope s for every h. What is left is rounding error, which grows as h gets tiny, because two almost equal losses are subtracted (see the Extra on central differences in what the sign of the derivative tells us).

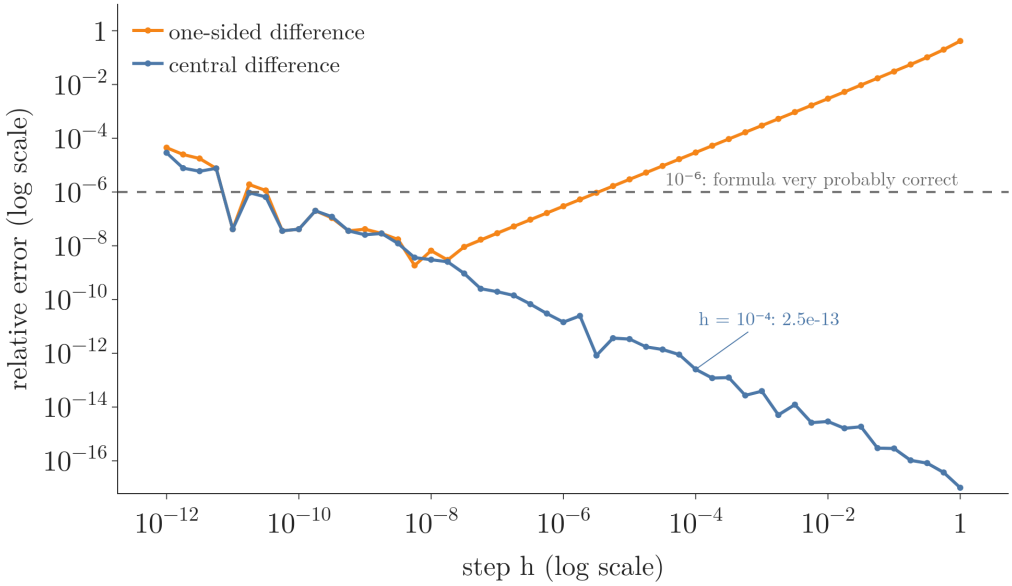


Figure 12: Relative error of the numerical gradient against the step h, on the three-point loss at m = 1, b = 0. Orange: one-sided difference, error shrinking with h until rounding takes over. Blue: central difference, exact for this quadratic loss except for rounding, which grows as h shrinks. Dashed: the 10^{-6} threshold.

Deep learning libraries compute gradients automatically, and they ship the same test for checking them: PyTorch's `torch.autograd.gradcheck` compares the automatic gradients with finite differences (PyTorch docs, "Gradcheck mechanics").

8. Summary

Object	Shape	Example for $f = x_1^2 + x_1x_2 + 2x_2^2$ at $(1, 1)$	Why it matters
Function value $f(\mathbf{x})$	number	4	the height of the surface, as a loss is one error
Partial derivative $\partial f/\partial x_i$	number	3 and 5	the slope along one input's axis
Gradient ∇f	$1 \times n$ row	$[3, 5]$	points steepest uphill; gradient descent steps against it
Chain rule $\partial f/\partial(s, t)$	$(1 \times 2)(2 \times 2) = 1 \times 2$	$[30, 19]$ at $s = 1$, $t = 2$	the gradient when f depends on its inputs through other variables

- A **partial derivative** (G-1457) moves one input and holds the others fixed; it is the slope of one slice of the surface, so the one-variable rules compute it, with the other inputs treated as constants.
- The gradient collects all partial derivatives; it points straight uphill, at right angles to the contour lines, and **gradient descent** (G-862) steps against it, because the slope along a unit direction $\mathbf{u}$ is $\nabla f \cdot \mathbf{u}$, largest when $\mathbf{u}$ points along the **gradient** (G-863).
- Sum, product and **chain rules** (G-371) carry over to vectors, with the order of factors kept, because the factors are now vectors and matrices.
- The **multivariate chain rule** (G-1281) adds the products along every path, because the input changes f through each intermediate variable; with gradients as rows, it is matrix multiplication.
- **Gradient checking** (G-860) compares a formula with finite differences, because a wrong hand-derived gradient makes training fail quietly.
- So the partial derivatives of a loss, collected in the gradient, tell gradient descent which way is steepest downhill.

9. Sources

Built from

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 5.2, remark on verifying a gradient implementation (MML).
- Khan Academy, "Partial derivatives, introduction", YouTube, https://www.youtube.com/watch?v=AX (Figure 4)
- Khan Academy, "Partial derivatives and graphs", YouTube, https://www.youtube.com/watch?v=dfvnC
- Khan Academy, "Gradient", YouTube, https://www.youtube.com/watch?v=tIpKfDc295M

- Khan Academy, “Gradient and graphs”, YouTube, https://www.youtube.com/watch?v=_-02ze7tf08 (Figure 6)
- Khan Academy, “Gradient and contour maps”, YouTube, https://www.youtube.com/watch?v=ZTbTY (Figure 7)
- Khan Academy, “Directional derivative”, YouTube, https://www.youtube.com/watch?v=N_ZRcLheN
- Khan Academy, “Why the gradient is the direction of steepest ascent”, YouTube, https://www.youtube.com/watch?v=TEB2z7ZlRAw (Figure 8 recreates its idea with our own function and code)
- Khan Academy, “Multivariable chain rule”, YouTube, https://www.youtube.com/watch?v=NO3AqAaA (Figure 11)
- Khan Academy, “Multivariable chain rule intuition”, YouTube, https://www.youtube.com/watch?v=hJx28 (Figure 11: a nudge in t splits into an x part and a y part)

Other references

- PyTorch documentation. “Gradcheck mechanics” and `torch.autograd.gradcheck`.

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Contour plot (G-468)	A map of a surface seen from above, with lines joining points of equal height, so a 3D shape such as a loss bowl can be drawn and read on flat paper.
Function of several variables (G-815)	A function that takes several numbers in (a vector of n inputs) and gives one number out, $f : \mathbb{R}^n \to \mathbb{R}$; a loss that depends on many weights is one.
Nabla (∇) (G-1295)	The upside-down triangle ∇, read “nabla” or “grad”; ∇f means the gradient of f, the vector that collects all its partial derivatives.
Gradient as a row vector (G-858)	Writing the gradient as one row of partial derivatives, $\nabla f = [\partial f/\partial x_1, \ldots, \partial f/\partial x_n] \in \mathbb{R}^{1\times n}$; this convention makes the chain rule a matrix product.
Directional derivative (G-614)	How steeply a function rises when we step in a chosen direction, given by a unit vector $\mathbf{u}$, instead of only along an axis. It is the dot product $\nabla f \cdot \mathbf{u}$.

Term	Meaning
Steepest ascent (G-1888)	The direction in which f increases fastest: the direction of the gradient; gradient descent steps the opposite way to go downhill fastest.
Multivariate chain rule (G-1281)	How to take a derivative when a function depends on its inputs through in-between (intermediate) variables: multiply the derivatives along each path and add the paths. In matrix form, a row gradient times a matrix of inner derivatives.
Gradient checking (G-860)	Checking that a gradient formula is right by comparing it with a numerical estimate made by nudging each weight a tiny amount (finite differences), measured by the relative error.
Partial derivative (G-1457)	The slope of a function of several variables in one variable, holding the others fixed.
Gradient (of the loss) (G-863)	How fast the loss changes with each weight and bias, collected in one vector (the partial derivatives of the loss); it points in the direction in which the loss rises fastest.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.
Jacobian (G-980)	The table (matrix) of all first partial derivatives of a function with several inputs and outputs, one row per output and one column per input, $J_{ij} = \partial f_i / \partial x_j$; it shows how every output changes with every input near a point.
Composition (G-431)	Applying one transformation and then another, seen as one overall transformation.